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ANDERSON JUNIOR COLLEGE
JC2 Preliminary Examination 2009

MATHEMATICS

Higher 3

Paper 1

9810/01

23 September

3 hours

Additional Materials : Answer Paper
Graph Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name and Class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions in Section A and any **four** questions from Section B.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers. Up to 4 marks will be awarded for the style and clarity of your mathematical presentation, based on the whole paper.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

	Section A				Section B (Please fill in the question numbers of the questions you attempted.)				Mathematical Presentation		
Qn. No:	1	2	3	4					MP1	MP2	Total
Marks											

This document consists of 6 printed pages (inclusive of this cover page).

Section A

Answer **all** the questions in this section

1. A differential equation is given by

$$\frac{dy}{dx} = (y^2 - a)(y^2 - 4) \text{ where } a \text{ is a parameter}$$

- (i) Locate the bifurcation values(s) . [4]
- (ii) Draw the bifurcation diagram for the parameter a , showing the phase lines and equilibrium values. [5]
- (iii) Comment on the stability of the equilibrium values for $a=0$ [2]
- (iv) For the case where $a = 16$, sketch on the same diagram the two solution curves passing through the points $(0,2)$ and $(0,3)$ respectively . [2]

2. Consider the initial value problem

$$\frac{dy}{dt} = \frac{1}{1-y}, \quad y(0) = 2$$

- (i) Apply the Improved Euler Method with a step size of 0.5 to estimate the value of y when $t = 1$. [3]
- (ii) By solving analytically, investigate the value of y at $t = 1$ and sketch the solution curve . [4]
- (iii) Comment on the Improved Euler Method in (i). [1]

3. The mean height $h(t)$ of a certain species of animals is governed by the differential equation

$$\frac{dh}{dt} = bh(1 - ah), \quad h(0) = h_0$$

where a and b are positive constants.

- (i) Using the substitution $h = \frac{1}{y}$, transform the differential equation into one containing y , and show that the resulting equation has the solution

$$y = a + (y_0 - a)e^{-bt}, \quad \text{where } y_0 = \frac{1}{h_0}, \quad h_0 < \frac{1}{a} \quad [5]$$

- (ii) Given that the mean height of these animals at birth is 0.4 m, find an expression for $h(t)$ in terms of a and b . [2]

- (iii) Given further that $a = 0.57$, find the maximum height of this group of animals. [2]

4. A particular population $P(t)$ of fish in a habitat satisfies the differential equation

$$\frac{dP}{dt} = 0.3P - 0.00012P^2 \quad \text{where } t \text{ is time in weeks.}$$

- (i) State the growth rate coefficient. [1]
- (ii) What is the limiting population and suggest one measure to increase the limiting population. [2]
- (iii) If the fish in a particular week is doubled due to emigration of the same species of fish from another region, how will the carrying capacity be affected? Explain your answer. [2]

The authority decides to let people fish in the habitat at a fixed rate h per week.

- (iv) For what values of h would extinction occur regardless of the initial population? [3]
- (v) If the initial population is 600, what is the largest value of h to ensure the continual survival of the fish population? [2]

Section B

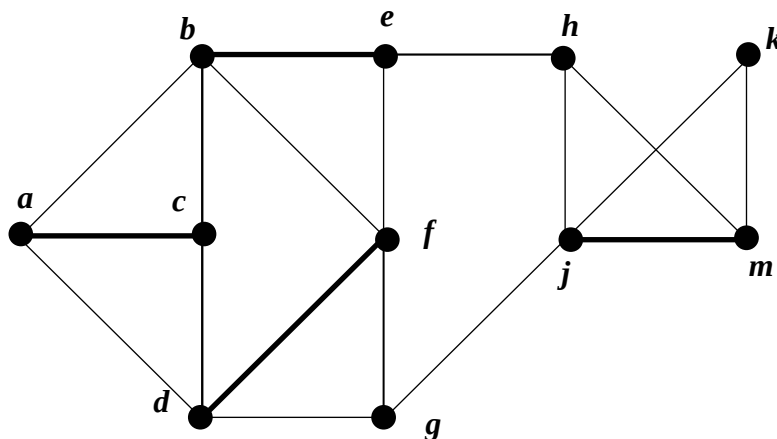
Answer any **four** questions from this section. Each question in this section carries 14 marks.

Plane Geometry

5. (a) In $\triangle ABC$, the bisector of $\angle A$ intersects BC at D . A perpendicular to AD from B intersects AD at E . A line segment through E and parallel to AC intersects BC at G , and AB at H .
- (i) Prove that H is the mid-point of AB . [5]
- (ii) If $AB = 26$, $BC = 28$, $AC = 30$, find the length of DG . [3]
- (b) In a right triangle $\triangle ABC$, AC is the hypotenuse. P is a point on AB such that $BP = PA = 2$. Point Q is a point on AC such that PQ is perpendicular to AC .
- (i) Prove that $BPQC$ is a cyclic quadrilateral. [1]
- (ii) If $CB = 3$, find the length of BQ . [5]
6. (a) In $\triangle ABC$, where CD is the altitude to AB and P is any point on DC , AP meets CB at Q , and BP meets CA at R . Prove that $\angle RDC = \angle QDC$, using Ceva's Theorem. [5]
- (b) A circle inscribed in $\triangle ABC$ is tangent to sides BC , CA and AB at points L , M and N respectively. If MN extended meets CB at P , prove that
- (i) $\frac{BL}{LC} = -\frac{BP}{PC}$, [4]
- (ii) if NL meets AC at Q and ML meets AB at R , then P , Q and R are collinear. [5]

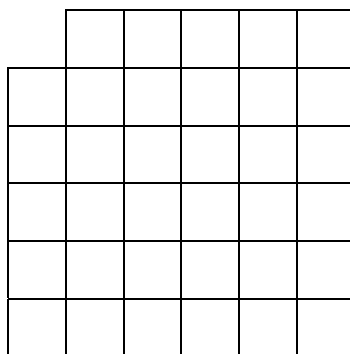
Graph Theory

- 7 (a) In the graph below, the marked edges form a matching. Show that this is a maximum matching, or if it is not, find a maximum matching. [3]



- (b) Let H be a bipartite graph with bipartition (X, Y) . Assume that $e(H) = 2v(H)$, and that $d(x) \geq 3$ for each x in X . Show that $|X| \leq 2|Y|$. Construct one such graph H with $|X| = 2|Y|$. [5]

- (c) Consider the following 6×6 grid-board whose upper left and lower right corner squares are removed. You are given 17 dominoes, each covering exactly two adjacent squares (squares that have an edge in common) of the board. Can you use them to cover the 34 squares in the board? [6]



- 8 (a) Let G be a graph of order $n \geq 2$ such that the minimum degree is $\frac{1}{2}(n-1)$. Show that the distance for any two vertices u, v in G , $d(u, v) \leq 2$. [5]
- (b) (i) Let G be a graph of order n such and size $m > \binom{n-1}{2}$. Show that G is connected. [7]
- (ii) For any $n \geq 2$, construct a disconnected graph of order n and size $\binom{n-1}{2}$. [2]

Combinatorics

9. The Stirling numbers of the second kind, denoted by $S(r, n)$, is defined to be the number of ways to distribute r distinct objects into n identical boxes such that no box is empty.
- i) Show that $S(r, n) = S(r-1, n-1) + nS(r-1, n)$ [3]
 - ii) Assuming the result in (i), prove by mathematical induction $S(r, 2) = 2^{r-1} - 1, r \geq 2, r \in \mathbb{N}$ [3]
 - iii) State what does $2!S(r, 2)$ represent, in terms of objects and boxes. [1]
 - iv) A promoter at a scholarship fair decided to issue highlighters to students who visit his booth. There are 5 different colours of highlighters for him to choose from and the highlighters are identical except for colours. A total of 4 students visited his booth on the first day of the event. How many ways can he issue the highlighters to each of the students if he uses at most 3 different colours of highlighters, with each student getting only 1 highlighter? It can be assumed that he has enough supply of highlighters for each colour. [5]
 - v) On the second day of the fair, 6 students visited his booth. As the promoter has an excess of yellow highlighters, he decided to issue a total of 15 yellow highlighters to the students, with each student getting at least one highlighter. How many ways can he do it? [2]
10. a) A special n -digit password used by PUSB bank can only contain the digits 0, 1 and 2. In order for the password to be valid, it must contain an odd number of zeros. Let V_n denote the number of such valid n -digit passwords. By considering the number of valid and invalid 8-digit passwords, show that $V_9 = 3^8 + V_8$. [5]
- b) Suppose a customer chooses to use a combination of the digits 0, 0, 0, 1, 1, 1, 2, 2, 2 as his password. Using the principle of inclusion and exclusion, find the number of ways he can arrange these nine digits so that no three consecutive digits are the same. [5]
- c) Five customers went to set up their accounts with the bank and were each issued a random queue number which is a positive integer. Show that among these five distinct queue numbers, there are three whose sum is divisible by 3. [4]

END OF PAPER